

Multiple Choice

1. **Answer:** A

Options: A) 0.3 liters; B) 2 liters; C) 0.2 liters; D) 3 liters; E) 0.03 liters

Note: The correct answer is 0.3 liters because, according to the principles of neutralization, the volume of base required is determined by the formula $N_1V_1 = N_2V_2$, where the normality (N) and volume (V) of the acid and base must be equal at the point of neutralization, leading to $0.3\text{ N} \times V_1 = 0.01\text{ N} \times 3\text{ L}$, resulting in $V_1 = 0.3\text{ L}$.

2. **Answer:** B

Options: A) 75.0 mL; B) 90.0 mL; C) 60.0 mL; D) 120 mL; E) 30.0 mL

Note: To neutralize 50.0 mL of 0.150 M H_3PO_4 , which has three acidic protons, a stoichiometric calculation shows that 90.0 mL of 0.250 M KOH is required to provide the necessary equivalent of hydroxide ions for complete neutralization.

3. **Answer:** D

Options: A) 0.1645 amu; B) 0.1987 amu; C) 0.1312 amu; D) 0.1588 amu; E) 0.1756 amu

Note: The correct answer of 0.1588 amu is obtained by calculating the difference between the total mass of the constituent protons and neutrons in the fluorine-19 nucleus and the actual atomic weight, accounting for the binding energy that results in the mass defect.

4. **Answer:** E

Options: A) 22,102.28 liters air; B) 58,650.00 liters air; C) 49,350.25 liters air; D) 65,292.15 liters air; E) 39,175.40 liters air

Note: The correct answer of 39,175.40 liters of air is derived from stoichiometric calculations based on the complete combustion of octane (C_8H_{18}), which requires approximately 25 moles of oxygen from the air for every mole of octane, and considering that air is approximately 21% oxygen by volume, leading to the final volume requirement at standard temperature and pressure (S.T.P).

5. **Answer:** B

Options: A) 2,3-dimethylbutane < neopentane < n-hexane < n-undecane < methane; B) methane < 2,3-dimethylbutane < neopentane < n-hexane < n-undecane; C) n-undecane < n-hexane < neopentane < 2,3-dimethylbutane < methane; D) neopentane < 2,3-dimethylbutane < methane < n-undecane < n-hexane; E) 2,3-dimethylbutane < methane < n-hexane < neopentane < n-undecane

Note: The boiling points of hydrocarbons increase with molecular weight and branching; thus, methane, being the smallest and least branched, has the lowest boiling point, while n-undecane, being the largest and unbranched, has the highest boiling point, with 2,3-dimethylbutane and neopentane having intermediate boiling points due to

their branching reducing their surface area and thus their boiling points relative to n-hexane.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: The statement is true because the entropy change of $16.2 \text{ J mol}^{-1} \text{ K}^{-1}$ for the vaporization of water indicates a significant increase in disorder as water transitions from a structured liquid phase to a more disordered gaseous phase, reflecting the energy distribution and molecular freedom associated with this phase change.

7. **Answer:** False

Options: A) True; B) False

Note: The maximum wavelength of the earth's blackbody radiation, determined by Wien's displacement law, is approximately $10.1 \text{ micrometers } (1.01 \times 10^{-5} \text{ m})$ at a mean surface temperature of 288 K .

7. **Answer:** True

Options: A) True; B) False

Note: The ammonium ion (NH_4^+) has a full octet around the central nitrogen atom because it is bonded to four hydrogen atoms, sharing a total of eight electrons in its valence shell.

8. **Answer:** True

Options: A) True; B) False

Note: The statement is true because the rate of gas diffusion is inversely proportional to the square root of its molar mass, and since hydrogen has a significantly lower molar mass than oxygen, it diffuses faster.

9. **Answer:** True

Options: A) True; B) False

Note: The statement is true because the molarity of the H_2SO_4 solution can be accurately determined through a neutralization reaction with NaOH, where stoichiometric calculations based on the balanced chemical equation allow for the determination of the acid's concentration.

Long Form Response

11. **Answer:** To calculate the molecular weight of the newly discovered compound based on the observed freezing point depression of water, we first apply the freezing point depression formula: $\Delta T_f = K_f \times m$, where ΔT_f is the change in freezing point, K_f is the freezing point depression constant for water ($1.86^\circ\text{C}/\text{mol}$), and m is the molality of the solution. The observed ΔT_f is 1.22°C , indicating that $\Delta T_f = 1.22$. Rearranging the equation to find molality gives $m = \Delta T_f / K_f = 1.22 / 1.86^\circ\text{C}/\text{mol} = 0.656 \text{ mol}/\text{kg}$. Since the mass of the solute is 2.00 grams , we convert this to kilograms (0.002 kg) and use the molality to find the molecular weight: $\text{MW} = \text{mass} / (m \times \text{kg}) = 2.00 \text{ g} / (0.656 \text{ mol}/\text{kg} \times 0.002 \text{ kg}) = 152.4 \text{ g}/\text{mol}$.

moles of solute/kg of solvent), we determine the number of moles of the compound : moles = $0.656 \text{ mol/kg} \times 0.002 \text{ kg} = 0.001312 \text{ mol}$. Finally, the molecular weight (MW) can be calculated using the formula $\text{mass of solute} / \text{moles of solute} = 2.00 \text{ g} / 0.001312 \text{ mol}$, resulting in a molecular weight of approximately 1525 g/mol. Note: correct because it accurately applies the freezing point depression formula to determine the molality.

11. **Answer:** To calculate the number of molecules in $1.1 \times 10^8 \text{ kg}$ of phosgene gas (COCl_2), we first convert the mass to moles using the molar mass of phosgene, which is approximately 98.92 g/mol . This results in approximately $1.11 \times 10^9 \text{ moles}$ of phosgene. By applying Avogadro's number ($6.022 \times 10^{23} \text{ molecules/mol}$), we find that this quantity corresponds to approximately 7.3×10^{31} molecules. This calculation is significant in the context of chemical warfare, as it illustrates the vast scale at which chemical agents like phosgene can be deployed, emphasizing their potential for mass casualty and the ethical implications of their use in warfare.
Note: correct because it accurately converts the mass of phosgene gas to moles using its molar mass and then applies Avogadro's number to determine the total number of molecules, highlighting the immense scale of chemical warfare capabilities.

End of Answer Key